

Python 1: Core Data Analysis - Assignments

Python 1 – Part 1

Exercise 1

In this exercise you will practice creating variables and learn the behaviour of casting.

1. Create the following variable(s):
 - a. Assign 4.3 to j.
 - b. Assign 5.5 to k.
2. Calculate $j * k$ and cast the result as an integer to a variable 'i'.
 - a. To cast to integer use `int()`.
3. Print the value and type of 'i'.
4. Print the value and type of 'j*k'.

What do you notice about the behavior of casting a float to an integer?

Exercise 2

In this exercise you will practice how to clean a numeric value that is stored as a string formatted for display purposes.

1. Assign '\$123,456,789' to a variable name 'deal_value'.
2. Use `.strip()` to remove the '\$'.
 - a. Remember to use " or " around the \$.
3. Use `.split()` to create a list divided at the commas.
4. Use `.join()` to create a string with no spaces in between.
5. Make the number useable (cast either to integer or float).
 - a. Check that you can multiply the number by 2.

Exercise 3

In this exercise you will practice looping through values in a list and using if statements to control execution.

1. Create the following list: [100, 99, 105, 110, 95, 102] and store the values in variable named 'mylist'.
2. Loop through the elements in the list and perform the following inside the loop:
 - a. If > 102 print "Sell"
 - b. If < 100 print "Buy"
 - c. Else print "Hold"

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Exercise 4

In this exercise you will practice using dictionaries and how to structure if statements effectively. Create a script that converts number grades to letter grades.

1. Create the following dictionary {'Bob':71, 'Alice':65, 'Jim':70, 'Jen':90, 'Tim':86, 'Trish':85, 'Tony':75} and store the values in a variable named 'grades'.
2. Loop through the keys in the 'grades' dictionary and convert each grade into letters following the rules below. Store the letter grades in a new dictionary.
Hint: create an empty dictionary named 'letter_grades' before the loop (letter_grades = {}).
Hint: use if/elif/else statement in a clever order so that you only need one logical check per if.
 - a. 90 and above: A+
 - b. 85-90: A
 - c. 80-85: A-
 - d. 75-80: B+
 - e. 70-75: B
 - f. 65-70: B-
3. Print the new dictionary with letter grades and double check that the letter grades correspond with the proper number grades in the original dictionary.

Exercise 5

In this exercise you will practice removing items from a list.

1. Create the following list: mylist = ['A', 'B', 'C', 'D', 'D', 'D'].
2. Create a loop using the syntax while 'D' in mylist, that will remove all instances of 'D' from the list.
 - a. Since .remove() will only remove the first instance of 'D', we need to use a loop to remove all instances of 'D' from the list.

Exercise 6

For this exercise, we will repeat Exercise 2, but instead of casting the whole string as a number, we will cast each individual chunk then combine them by adding the numbers (e.g. 123,000,000 + 456,000 + 789). You will practice using for loops using a range, and programming logic.

1. Use .split() to get a list ['123', '456', '789'].
2. Use a for loop to iterate over the range 0 to 2.
Hint: The function len() will get you the number of elements.
Hint: Initialize the variable you will store the sum before the loop.
3. Inside the loop, cast each number to an integer, and multiply by 1,000^(number of elements – index - 1) to get the proper magnitude.
Hint: To access the values stored in the list, use the loop index: mylist[i].

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Advanced Questions

Exercise 7

In this exercise you will practice using functions, working with lists & dictionaries and formatting output. The function will simulate random prices movements of stocks.

1. Create a dictionary with the following {'AAPL':100.0, 'CAT':50.0, 'MSFT':150}. These will be the starting prices for tickers AAPL, CAT, MSFT.
2. Create a function that takes the dictionary as defined above and generates 5 random prices movements using the standard normal distribution. Store each of the movements in a dictionary and return this dictionary.
Hint: Remember that you can only store a single value for each key, you will need to use a list to store the five prices movements ({key:[list]}).
3. The output should be formatted to display the ticker name and then the five price movements, to two (2) decimal places, in one line.
e.g.:
AAPL: \$99.44, \$98.99, \$100.31, \$99.09, \$98.59
CAT: \$51.47, \$49.77, \$50.07, \$48.58, \$49.46
MSFT: \$150.11, \$148.85, \$150.38, \$149.40, \$149.71
AAPL \$100, CAT \$50, MSFT \$150

Exercise 8

In this exercise you will practice using matrices and generating random numbers.

1. Use a nested loop to generate a matrix of simulated monthly returns with means 1 to 5 and standard deviation 1 to 5 for each.
 - a. 25 total assets, 60 returns for each
 - b. Have each column be an asset, and rows be the return observations (this will result in a 60x25 matrix).
 - c. Each asset has a different combination of mean/standard deviation for monthly returns (e.g. the first 5 assets will have means of 1, and standard deviation of 1 to 5, the next 5 assets will have means of 2, etc.)
 - d. We will assume IID, so each draw is unique and not dependent on the previous draws.
2. Find the covariance matrix and store the result in a variable, using `np.cov(matrix)`.
3. Find the inverse of the covariance matrix, if possible, and verify the inverse is correct (Remember that $A^{-1}A = I$).

Challenge: Verify that the covariance matrix is positive definite.

Hint: The Cholesky Decomposition $A = LL'$ only holds for positive definite matrices.